

HOORN-A, Netherlands

WMO: 062120

Lat: 52.918N

Lon: 4.150E

Elev: 135

StdP: 14.62

Time Zone: 1.00 (EUC)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	29.8	32.0	19.4	14.7	31.6	22.6	17.1	33.0	45.7	46.8	42.5	46.4	23.5	80	0.829

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.0	71.8	65.4	69.1	64.2	67.2	63.0	66.6	70.9	65.1	68.4	63.7	66.5	15.1	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.0	92.8	68.8	63.5	88.2	67.4	62.2	84.2	66.1	31.3	71.1	30.2	68.7	29.1	66.5	75.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
40.1	36.4	32.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.9	42.3	41.5	43.1	47.9	52.4	57.3	62.1	63.4	61.0	56.2	49.4	45.3	(6)
	DBStd	8.76	4.60	4.27	4.37	4.60	4.56	3.81	3.79	3.38	3.45	3.83	4.51	4.97	(7)
	HDD50	1026	240	237	216	91	24	0	0	0	0	4	61	153	(8)
	HDD65	4835	705	657	679	513	392	235	107	69	128	272	467	611	(9)
	CDD50	1715	0	0	2	28	97	219	375	415	331	197	44	7	(10)
	CDD65	47	0	0	0	0	0	3	17	19	8	0	0	0	(11)
	CDH74	69	0	0	0	0	0	9	37	13	10	0	0	0	(12)
	CDH80	7	0	0	0	0	0	0	7	0	0	0	0	0	(13)
(14) Wind	WSAvg	17.6	21.0	20.3	18.6	16.0	16.0	14.8	14.2	14.8	15.8	18.9	19.5	21.5	(14)
(15) Precipitation	PrecAvg	35.5	3.0	2.5	2.0	1.7	2.2	2.6	3.2	3.9	3.5	3.7	3.8	3.4	(15)
	PrecMax	46.8	5.6	5.1	4.6	3.9	3.9	6.0	6.8	8.9	7.7	7.0	6.9	6.2	(16)
	PrecMin	26.0	0.3	0.8	0.4	0.0	0.8	1.0	0.8	0.9	0.6	0.8	0.4	0.9	(17)
	PrecStd	5.0	1.4	1.1	1.1	0.9	0.9	1.4	1.6	2.2	1.8	1.6	1.5	1.4	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.3	51.3	54.9	64.3	68.6	74.4	79.7	75.9	75.0	65.9	60.2	54.8	(19)
		MCWB	48.9	46.2	49.1	53.9	57.5	65.5	68.2	66.9	67.9	61.5	57.5	52.2	(20)
	2%	DB	50.2	49.2	50.9	59.5	63.2	67.0	71.1	71.0	69.0	63.8	58.5	53.7	(21)
		MCWB	47.8	45.7	47.3	52.5	56.7	62.8	65.3	65.0	64.8	60.2	55.8	51.4	(22)
	5%	DB	49.0	47.9	49.1	55.8	60.1	63.5	68.5	68.7	66.7	62.2	57.3	52.5	(23)
		MCWB	46.7	45.1	46.2	50.6	55.7	60.3	64.3	64.0	62.9	58.9	54.5	50.2	(24)
	10%	DB	47.9	46.7	47.7	53.3	57.8	61.4	66.3	67.0	65.2	61.1	55.5	51.1	(25)
		MCWB	45.5	44.3	45.2	49.6	54.7	58.4	62.4	62.8	61.3	57.8	52.2	48.6	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.7	48.5	50.0	55.0	60.2	65.9	68.3	67.9	69.1	62.1	58.5	53.1	(27)
		MCDB	50.9	49.7	54.2	62.2	64.5	71.2	74.1	73.1	74.2	64.8	59.7	54.1	(28)
	2%	WB	48.4	46.6	48.1	52.9	57.8	63.3	66.3	66.2	66.1	60.6	56.5	52.0	(29)
		MCDB	49.8	48.1	50.3	58.1	61.3	66.8	70.9	70.0	69.2	63.0	58.3	53.3	(30)
	5%	WB	47.2	45.6	46.9	51.3	56.2	60.8	64.8	64.7	63.7	59.5	54.8	50.7	(31)
		MCDB	48.8	47.3	48.7	54.8	59.5	63.1	68.0	67.7	66.1	61.9	57.1	52.4	(32)
	10%	WB	46.0	44.7	45.7	50.1	54.9	58.8	63.0	63.6	62.2	58.4	52.6	49.0	(33)
		MCDB	47.8	46.5	47.4	53.0	57.9	60.8	65.5	66.6	64.9	60.9	55.0	50.9	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	4.3	3.8	4.3	4.9	4.3	4.0	4.1	4.0	3.8	4.4	4.4	4.3	(35)
		MCDBR	4.7	4.3	6.0	9.1	8.9	8.5	8.1	7.1	5.2	4.9	5.1	4.0	(36)
	5% WB	MCWBR	5.4	4.0	4.4	5.3	5.3	5.9	5.4	4.7	5.1	4.8	5.7	5.1	(37)
		MCWBR	4.7	4.0	5.6	8.1	7.4	8.3	7.5	5.9	4.8	4.9	5.0	4.0	(38)
(40) Clear-Sky Solar Irradiance	taub		0.334	0.350	0.379	0.402	0.402	0.401	0.409	0.401	0.385	0.366	0.343	0.325	(40)
		taud	2.327	2.355	2.304	2.285	2.354	2.369	2.356	2.391	2.426	2.414	2.360	2.320	(41)
	Ebn at Noon		202	238	254	263	269	270	266	261	251	229	196	181	(42)
		Edh at Noon	20	25	32	37	37	36	36	33	29	24	19	17	(43)
(44) All-Sky Solar Radiation	RadAvg		204	421	837	1374	1696	1781	1707	1427	1000	549	242	153	(44)
	RadStd		19	73	81	128	104	105	173	105	80	48	22	16	(45)

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page